

DEVELOPMENT OF MAGNET PROTOTYPE FOR SIAM PHOTON SOURCE II

S. Prawanta, P. Sunwong, T. Leetha, P. Numanoy, P. Pruekthaisong, M. Sroison
N. Thiabsi, P. Chaithawee, P. Klysubun
Synchrotron Light Research Institute, Nakhon Ratchasima, Thailand

Abstract

The Siam Photon Source II (SPS-II) is Thailand's next-generation 3 GeV synchrotron light source, designed to deliver high-brightness radiation for scientific and industrial use. Its storage ring, comprising 14 Double Triple Bend Achromat (DTBA) cells and targeting a natural beam emittance below 1 nm·rad, requires numerous high-precision magnets with strict mechanical and magnetic specifications. This work presents the development of magnet prototypes, covering the full design process including magnet type and material selection, yoke and pole geometry optimization, and magnetic field simulation with OPERA 3D considering saturation and permeability. Mechanical design involved core and coil material selection as well as structural, thermal, and vibrational analyses using AutoCAD, SolidWorks, and ANSYS to ensure mechanical integrity. Prototypes were fabricated using high-precision machining and vacuum pressure impregnation (VPI) for coil insulation, then tested for dimensional accuracy and magnetic field performance. This marks Thailand's first domestically developed magnet prototype, realized through collaboration with local industry, laying a solid foundation for the SPS-II project.

INTRODUCTION

The Siam Photon Source II is a fourth-generation synchrotron light source currently under development in Thailand [1]. The facility consists of a 150 MeV linear accelerator, a 3 GeV booster synchrotron, and a 3 GeV storage ring with a circumference of 327.6 m, as presented in Fig. 1. The storage ring comprises 14 Double Triple Bend Achromat (DTBA) cells and is designed to achieve a natural beam emittance below 1 nm·rad, providing high brightness at an RF frequency of 496.654 MHz with a beam current

of 300 mA. The stability of the electron beam and the quality of the synchrotron light depend on the overall performance of the storage ring. The magnets, which control the trajectory and focusing of the electron beam, must possess accurate magnetic field characteristics and be installed and aligned with high precision.

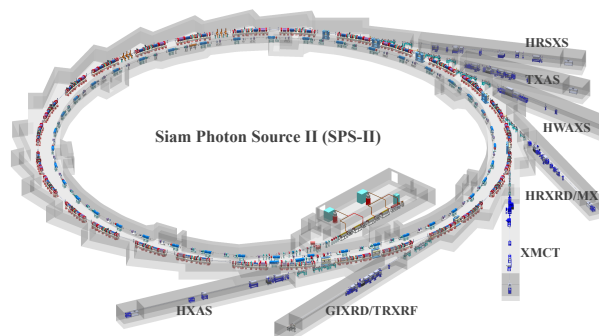


Figure 1: Overview of the SPS-II accelerator complex.

The development of a magnet prototype for the storage ring was carried out to produce a magnet with the required mechanical properties and precise magnetic field characteristics in accordance with its functional specifications. Key processes included magnetic design, mechanical design, electrical design, and prototype fabrication, serving as preliminary steps before entering full-scale production.

MAGNET DESIGN & ENGINEERING

The DTBA cell of the storage ring consists of various types of magnets, including dipole, quadrupole, dipole-quadrupole, sextupole, octupole, steering, and fast corrector magnets, as shown in Fig. 2 [2 - 4]. It can be seen that the magnets are mounted on three upstream girders and another three downstream girders.

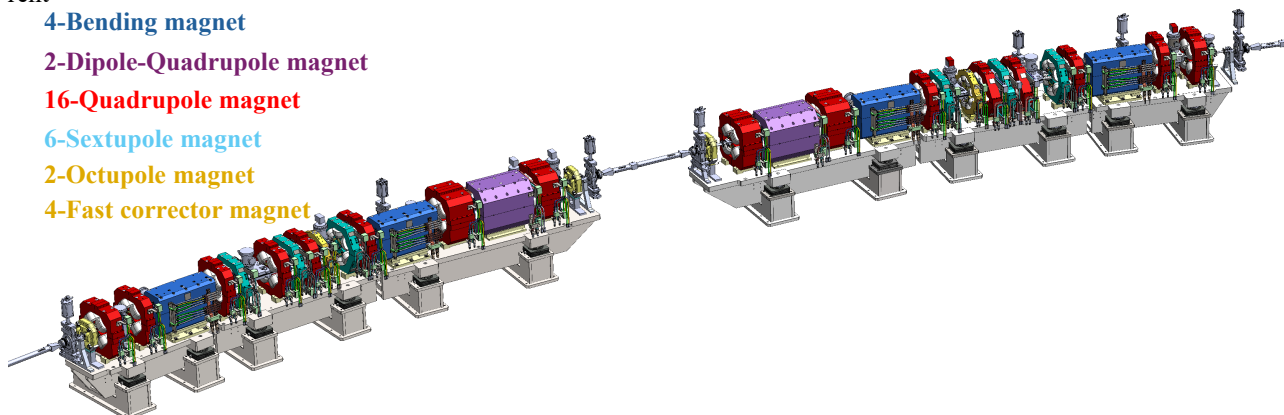


Figure 2: Double-Triple Bend Achromat (DTBA) cell of the storage ring.

The magnet design process encompasses magnetic, mechanical, and electrical aspects to ensure optimal performance under operational conditions. Magnetic design involves the selection of magnet types and materials, the design of yoke and pole geometries, and magnetic field simulation and optimization using OPERA 3D, taking into account magnetic saturation and material permeability. Mechanical design includes the selection of magnetic core and coil materials, as well as mechanical analyses—such as deformation, natural frequency, and thermal behavior—conducted using AutoCAD, SolidWorks, and ANSYS to verify structural integrity. For the electrical design, coil configuration, number of turns, conductor size, and insulation are specified. The current rating is calculated to meet power and thermal constraints, with water cooling integrated to maintain safe operating temperatures.

MAGNET PROTOTYPE FABRICATION

Magnet Yoke-Pole Fabrication involves the use of AISI 1006 low-carbon steel for the magnet yoke and pole pieces due to its high magnetic permeability. Ultrasonic testing is performed to detect internal flaws and ensure material homogeneity. Precision machining processes include CNC machining for high-accuracy geometries, surface grinding for smooth finishes and flatness control, and wire-cut electrical discharge machining (EDM) for precise pole contouring. These processes have a direct impact on the magnetic field quality in the good field region.

Dimensional inspection is performed using a coordinate measuring machine (CMM) to verify the dimensional accuracy of the assembled magnet yoke and pole pieces, ensuring that tolerances are maintained within $\pm 20 \mu\text{m}$.

The magnet is initially assembled and mounted onto the girder as illustrated in Fig. 3, followed by precise alignment. After alignment, the upper magnet yoke is removed to allow for vacuum chamber installation. The magnet is then reassembled and inspected to ensure compliance with design specifications.

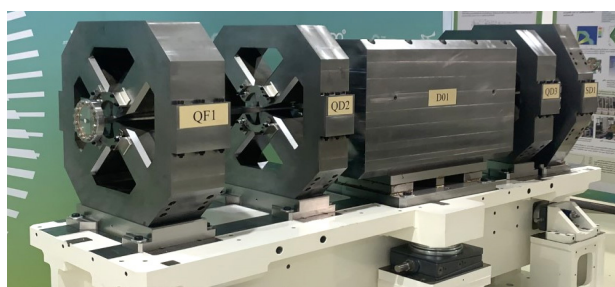


Figure 3: Assembly and integration.

Magnet coil fabrication uses OF-OK® oxygen-free copper hollow conductors, known for their high purity and excellent electrical and thermal conductivity. Meta-aramid fiber tape is applied in overlapping layers to ensure high electrical insulation and long-term reliability. The coil design optimizes current ratings, number of coil turns, and conductor size to reduce power loss and enhance heat dissipation.

A specialized winding machine has been developed for precise insulation application and coil winding according to specifications. This system includes a hollow conductor tensioner, an insulation tape wrapping unit, and a coil winding module, enabling accurate and efficient magnet coil fabrication.

Bending magnet coil winding, as shown in Fig. 4, presents specific challenges due to the large size of the hollow conductor, which complicates handling and bending. Additionally, half of the hollow conductor must be temporarily stored during the process before reversing the winding direction to form the second coil layer.

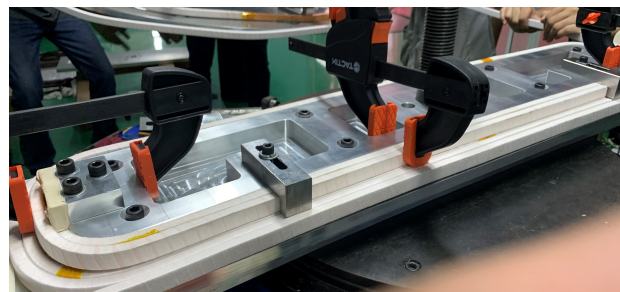


Figure 4: Bending magnet coil winding.

Vacuum pressure impregnation (VPI) is a specialized process developed to enhance coil performance by improving insulation, mechanical strength, and environmental protection. Epoxy resin penetrates internal gaps, creating a compact, void-free insulation layer. It also firmly bonds coil layers, minimizing movement or vibration that could cause damage. Additionally, the resin coating seals the coil, preventing moisture and dust ingress, thereby reducing the risk of short circuits. Figure 5 shows the magnet coil of dipole magnet before VPI insulation process and Fig. 6 are magnet coils after VPI process.

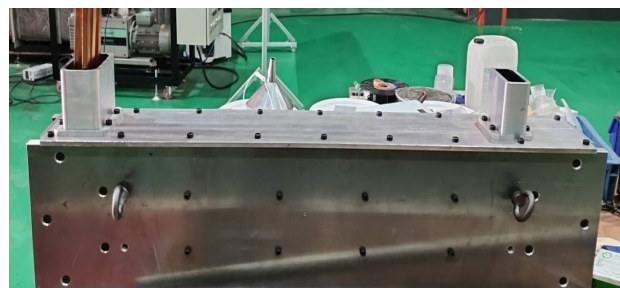


Figure 5: Magnet coil before VPI insulation process.



Figure 6: Magnet coils after VPI insulation process.

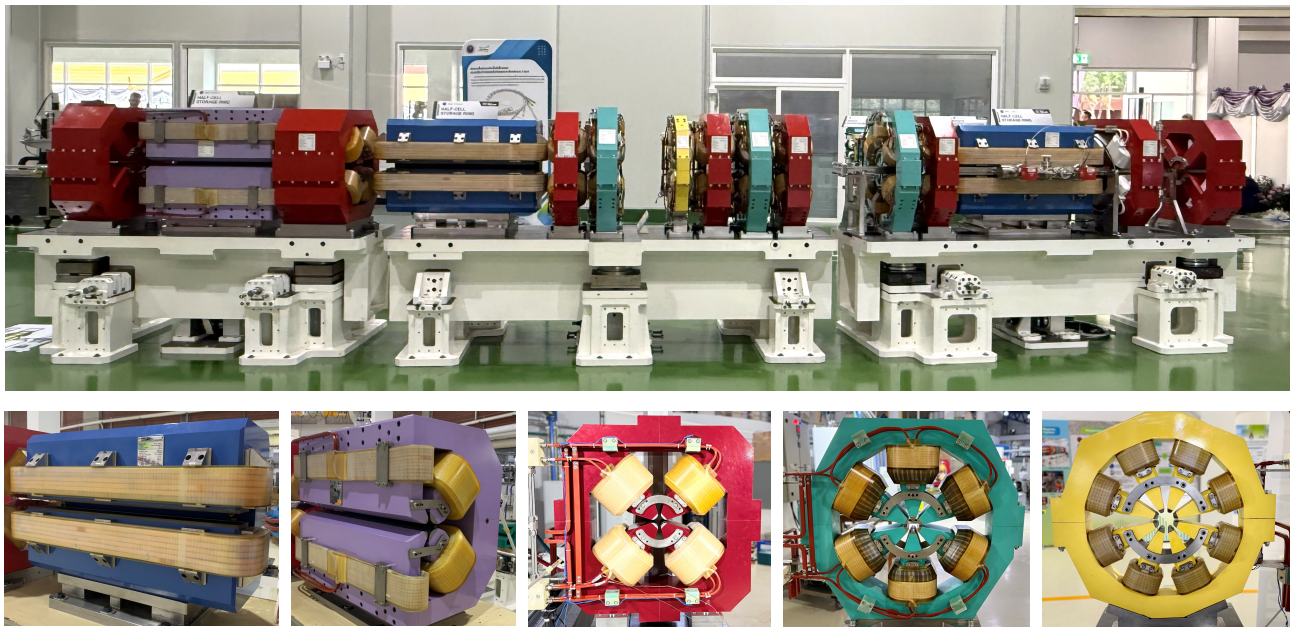


Figure 7: Fabrication of magnet prototype for the half-DTBA cell of the storage ring.

MAGNETIC FIELD MEASUREMENT

Magnet prototype for half-cell of DTBA lattice is illustrated in Fig. 7. Prototype testing involves specialized magnetic field measurement systems developed to validate magnetic design and simulations, ensure field quality and alignment, and optimize prototype performance prior to mass production.

The Hall probe measurement system measures the magnetic field distribution and strength of bending (BM), quadrupole (QM), sextupole (SM), and octupole (OM) magnets, which is essential for evaluating field quality and verifying magnetic length.

The stretched wire measurement system, as shown in Fig. 8, assesses the integrated field, multipole components, uniformity, symmetry, and center alignment of QM, SM, and OM magnets. This system is particularly effective for small-bore magnets and plays a critical role in field quality evaluation and alignment verification.

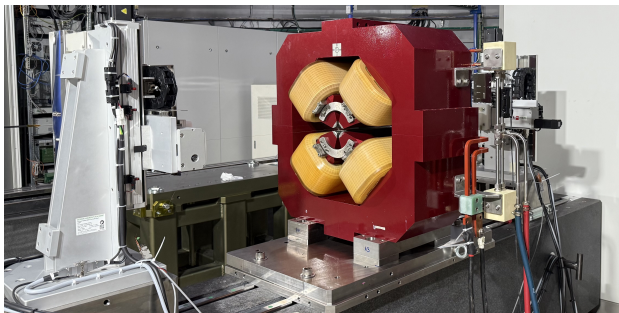


Figure 8: Stretched wire measurement system.

CONCLUSION

The development of magnet prototype for the SPS-II electron storage ring can be summarized as follows: The assembly and installation of the yoke and pole achieved

approximately 80% of the specified design precision and accuracy. Magnetic field measurements were consistent with simulation results, although the measured values have not yet fully met the design specifications. Nevertheless, the development of this prototype magnet marks a significant milestone in validating design concepts, manufacturing techniques, and alignment processes. The knowledge, technology, and experience gained from this work provide a solid foundation for the construction of the 3 GeV synchrotron light source and support future industrial-scale magnet production within the country.

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